

# MOYI TIAN

Applied Mathematics, 526 UCB, Boulder, CO, 80309, United States

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🌐 <https://moyi-tian.github.io/moyi-tian>

## EDUCATION

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**Brown University**, Providence, RI *September 2019 - May 2024*  
Ph.D in Applied Mathematics  
Thesis Advisor: Dr. Björn Sandstede, Division of Applied Mathematics  
Thesis: Patterns in Network Dynamics  
MS, Applied Mathematics *May 2021*  
**Dickinson College**, Carlisle, PA *September 2015 - May 2019*  
B.S. in Mathematics & Physics  
Phi Beta Kappa Honors, Honors in Mathematics, *Summa Cum Laude*

## POSITIONS

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### Academic

**University of Colorado Boulder**, Boulder, CO *2024 - Present*  
*Postdoctoral Associate*  
PI: Nancy Rodriguez (co-PI: David Bortz)

### Industry

**Los Alamos National Laboratory**, Los Alamos, NM *September 2023 - November 2023*  
*Applied Mathematics Graduate Student Intern*  
**SAS Institute, Inc.**, Remote *May 2023 - August 2023 (full-time)*  
*Machine Learning Graduate Intern* *& January 2024 - May 2024 (part-time)*  
**Oak Ridge National Laboratory**, Oak Ridge, TN *June 2022 - August 2022*  
*NSF Mathematical Sciences Graduate Intern*

## PUBLICATIONS

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( $\alpha - \beta$ ): authors listed alphabetically.

### Published Articles

- 2026 **M Tian**, PJ Brantingham, and N Rodríguez, *Modelling the Spillover from Online Engagement to Offline Protest: Stochastic Dynamics and Mean-Field Approximations on Networks*, Journal of Complex Networks **14** (2026), no. 2, cnaf057
- 2023 **M Tian** and P Moriano, *Robustness of Community Structure under Edge Addition*, Physical Review E **108** (2023), 054302
- 2021 **M Tian**, JJ Bramburger, and B Sandstede, *Snaking Bifurcations of Localized Patterns on Ring Lattices*, IMA Journal of Applied Mathematics (2021), hxab023
- 2019 H Mhiri, **M Tian**, E Wynne, S Jones, A Mareno, and LQ English, *An Experimental Survey of Chaos and Symmetry Breaking in Coupled and Driven Logistic Maps*, European Journal of Physics **40** (2019), no. 6, 065802

### Preprints

- 2026 **M Tian**, DA Messenger, V Dukic, N Rodríguez, and DM Bortz, *Learning Effective Models from Network Dynamics Data with Multiple Initial Conditions Using Weak Form SINDy*, arXiv:2605.30432 (2026)

- 2026     **M Tian**, G Mohler, PJ Brantingham, and N Rodríguez, *Learning Dynamics from Online-Offline Systems of LLM Agents*, arXiv:2602.23437 (2026)

## In Preparation

- **M Tian**, and B Sandstede, *Bifurcation from Homogeneous Branches on Ring Lattices*.
- $(\alpha - \beta)$  HZ Brooks, PS Chodrow, T Gebhart, L Huynh, V Modisette, W Thompson, **M Tian**, and A Wiedemann, *Learning Interaction Kernels in Asynchronous Opinion Dynamics on Networks*.

## SELECTED AWARDS, HONORS AND MEMBERSHIPS

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### Awards

- 2026     **Dynamics Days 2026 Travel Awards**, Dynamics Days US 2026  
Travel support to attend Dynamics Days US 2026 conference
- 2025     **Dynamics Days 2025 Travel Awards**, Dynamics Days US 2025  
Travel support to attend Dynamics Days US 2025 conference
- 2024     **SIAM Student Chapter Certificate of Recognition**, Society for Industrial and Applied Mathematics  
In recognition of outstanding efforts and accomplishments on behalf of the SIAM Chapter at Brown University
- 2023     **SIAM Student Travel Awards**, Society for Industrial and Applied Mathematics  
Support students to attend and present at SIAM conferences
- 2019     **James Fowler Rusling Prize**, Dickinson College  
Presented to a member of the senior class whose scholarly achievements have been judged most superior by the All-College Committee on Academic Program and Standards

### Honors

Phi Beta Kappa Honor Society  
Pi Mu Epsilon National Honorary Mathematics Society  
Sigma Pi Sigma National Physics Honor Society  
Alpha Lambda Delta Honor Society

### Memberships

- 2025 - Present    Association for Women in Mathematics
- 2019 - Present    Society for Industrial and Applied Mathematics
- 2019 - 2024       American Mathematical Society

## SELECTED PRESENTATIONS

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### Talks

- 2026     **APS Global Physics Summit 2026**  
*Learning Models from Network Dynamics Data using Weak Form SINDy*, Denver, CO, March, 2026
- 2026     **Dynamics Days US 2026**  
*Modeling and Learning Dynamics of Online-Offline Social Systems on Networks*, Tucson, AZ, January, 2026

- 2025 **ICERM Workshop on Patterns, Dynamics, and Data in Complex Systems**  
*Localized Patterns and Modeling Dynamics on Networks*, Providence, RI, January, 2025
- 2024 **2024 Joint Mathematics Meetings (JMM 2024)**  
*Inferring Interaction Kernels for Stochastic Agent-Based Opinion Dynamics*, San Francisco, CA, January, 2024
- 2023 **SIAM Conference on Applications of Dynamical Systems (DS23)**  
*Localized Patterns on Graphs*, Portland, OR, May, 2023
- 2022 **SIAM Workshop on Network Science (NS22)** (virtual) - Lightning Talk  
*How Robust are Communities in Temporal Networks? A Comparative Analysis Using Community Detection Algorithms*, September, 2022
- 2021 **Brown / BU / UMass Dynamics & PDE Seminar**  
*Snaking Bifurcations of Localized Patterns*, University of Massachusetts Amherst, Amherst, MA, November, 2021

## Posters

- 2025 **SIAM Conference on Applications of Dynamical Systems (DS25)**  
*Modeling Coupled Online and Offline Dynamics of Protesting Activity on Networks*, Denver, CO, May, 2025
- 2025 **Dynamics Days US 2025** - Flash Talk & Poster  
*Localized Patterns on Ring Lattices*, Denver, CO, January, 2025
- 2024 **AWM Workshop at SIAM Annual Meeting 2024**  
*Efficiently Learning Models of Networks*, Spokane, WA, July, 2024
- 2024 **2024 Dynamics Days US**  
*Efficient Learning of Models for Temporal Networks*, Davis, CA, January, 2024
- 2023 **Mathematical Opportunities in Digital Twins (MATH-DT) Workshop**  
*Community Robustness under Edge Addition in Synthetic and Empirical Temporal Networks*, George Mason University, Fairfax, VA, January, 2023
- 2023 **Dynamics Days US 2023** (virtual)  
*Community Robustness in Temporal Networks under Edge Addition*, January, 2023
- 2021 **SIAM Conference on Applications of Dynamical Systems (DS21)** (virtual)  
*Snaking Bifurcations of Localized Patterns on Ring Lattices*, May, 2021

## SELECTED WORKSHOPS / SUMMER SCHOOLS ATTENDED

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- 2025 IMSI Workshop: UQ and Trustworthy AI Algorithms for Complex Systems and Social Good, Chicago, IL, March 3 - 7, 2025
- 2024 IPAM Workshop IV: Modeling Multi-Scale Collective Intelligences, Los Angeles, CA, November 18 - 22, 2024
- 2023 American Mathematical Society Mathematics Research Communities 2023 on Complex Social Systems, Java Center, NY, June 18 - June 24, 2023
- 2023 Women in Network Science Collabathon, group project on topic modeling, Northeastern University, May 1 - May 5, 2023
- 2022 Fall 2022 Data Science Boot Camp, group project on English language proficiency evaluation model, the Erdős Institute, September - December, 2022 (virtual)
- 2022 OLCF Summer Hands-On High Performance Computing Course, Certificate of Completion, Oak Ridge National Laboratory, July 2022 (virtual)

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | May 2022 Data Science Boot Camp, group project on predicting chocolate ratings with feature engineering, the Erdős Institute, May 9 - June 4, 2022 (virtual) |
| 2020 | IMSD Module: Introduction to Statistical Analysis of Data 2020, Brown University, November 5, 12 and 19, 2020 (virtual)                                      |
| 2020 | IMSD Module: Scientific Presentations, Brown University, June 4, 5 and 11, 2020 (virtual)                                                                    |
| 2018 | Budapest Semesters in Mathematics Program, completed with honors, Budapest, Hungary, Summer 2018                                                             |

## LEADERSHIP AND SERVICE

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### Leadership

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|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fall 2021   | <b>Brown Division of Applied Mathematics - Directed Reading Program</b> , Brown University<br>Mentored an undergraduate student on mathematical optimal control theory and applications |
| Summer 2020 | <b>5-week Undergraduate Program in Experimental Math</b> , Brown University<br>Supervised a team of 3 undergraduate students on studying the dynamics and patterns of graphing fleas    |
| 2019 - 2020 | <b>Brown Applied Math Undergraduate/Graduate Mentoring Program</b><br>Mentored an undergraduate student with regards to study plans and academic/career goals                           |

### Service

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| 2021 - 2023 | <b>APMA Diversity, Equity, and Inclusion Committee</b> , Brown University<br>Attended bi-weekly meetings and developed plans and projects to improve climate and increase diversity and inclusion in the Division of Applied Mathematics    |
| 2021 - 2023 | <b>Brown SIAM Student Chapter Executive Board</b> , Brown University<br>Kept record for organizational plans and helped facilitate on-campus events                                                                                         |
| 2021 - 2023 | <b>Sheridan Center Departmental Graduate Student Liaison</b> , Brown University<br>Maintained communications and forwarded events information between the Sheridan Center for Teaching and Learning and the Division of Applied Mathematics |

## TEACHING

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### Experience

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|-------------|----------------------------------------------------------------------------------------------------------------|
| Fall 2025   | <b>Introduction to Differential Equations with Linear Algebra</b> , Instructor, University of Colorado Boulder |
| Spring 2021 | <b>Applied Ordinary Differential Equations</b> , Teaching Assistant, Brown University                          |
| Fall 2020   | <b>Operations Research: Deterministic Models</b> , Teaching Assistant, Brown University                        |
| Fall 2018   | <b>Single Variable Calculus</b> , Teaching Assistant, Dickinson College                                        |
| Fall 2017   | <b>Introduction to Calculus</b> , Teaching Assistant, Dickinson College                                        |
| Fall 2016   | <b>Single Variable Calculus</b> , Teaching Assistant, Dickinson College                                        |

## Pedagogical Training

Fall 2020      **Sheridan Center Certificate I: Reflective Teaching**, Brown University  
Introductory seminar in a cross-disciplinary setting that helps develop and refine  
fundamental, evidence-based teaching skills and strategies

## Tutoring

2017 - 2019      **Quantitative Reasoning Center**, Dickinson College  
Tutored college entry-level math, physics and economics students/classes

2016 - 2019      **Math Help Room**, Dickinson College  
Tutored walk-in students from college entry-level mathematics classes